

Kevin D. Ricks

Curriculum Vitae

Department of Ecology and Evolutionary Biology
University of Toronto

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EDUCATION

Ph.D. Ecology, Evolution, and Conservation Biology Advisor: Dr. Anthony Yannarell	University of Illinois at Urbana-Champaign	2023
M.S. Biology Advisor: Dr. Roger Koide	Brigham Young University	2018
B.S. Environmental Science (<i>cum laude</i>) Minor: Statistics	Brigham Young University	2016

PROFESSIONAL APPOINTMENTS

Arts and Science Postdoctoral Fellow Advisor: Dr. Megan Frederickson	University of Toronto	2023–Current
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PUBLICATIONS

- Ricks KD**, Raglin SS, & Kent AD (2025). Signature of plant nitrogen adaptation in the associated root-microbiome across genotype panel of *Brachypodium distachyon*. *New Phytologist*. doi: 10.1111/nph.70684
- Ricks KD**, Ricks NJ, & Yannarell AC (2023). Patterns of plant salinity adaptation depend on interactions with soil microbes. *The American Naturalist*. doi: 10.1086/725393
- Ricks KD** & Yannarell AC (2023). Soil moisture incidentally selects for microbes that facilitate locally adaptive plant response. *Proceedings of the Royal Society B*. doi: 10.1098/rspb.2023.0469
- Nettles R, **Ricks KD**, & Koide RT (2020). The dynamics of antibiotic-induced dysbiosis in interacting bacterial and fungal communities of the mouse colon. *Microbial Ecology*. doi: 10.1007/s00248-020-01525-6
- Ricks KD** & Koide RT (2019). The role of inoculum dispersal and plant species identity in the assembly of leaf endophytic fungal communities. *PLOS ONE*. doi: 10.1371/journal.pone.0219832
- Ricks KD** & Koide RT (2019). Biotic filtering of endophytic fungal communities in *Bromus tectorum*. *Oecologia*. doi: 10.1007/s00442-019-04388-y
- Koide RT, Nguyen BT, Skinner RH, Dell CJ, Adler PR, Drohan PJ, Peoples MS, Licht M, Boyer M, Nettles R, **Ricks KD**, & Watkins J (2018). Comparing biochar application methods for switchgrass yield and C sequestration on contrasting marginal lands in Pennsylvania, USA. *BioEnergy Research*. doi: 10.1007/s12155-018-9940-1

- Koide RT, **Ricks KD**, & Davis ER (2017). Climate and dispersal influence the structure of leaf fungal endophyte communities of *Quercus gambelii* in the eastern Great Basin, USA. Fungal Ecology. doi: 10.1016/j.funeco.2017.08.002
- Koide RT, Watkins J, **Ricks KD**, Aranda E, Nettles RM, Yokum H, Yin N, & Clark E (2017). Site and tree factors determining the distribution of *Phellinus tremulae* in *Populus tremuloides* in Utah, USA. Canadian Journal of Forest Research. doi: 10.1139/cjfr-2017-0323
- Nettles R, Watkins J, **Ricks KD**, Boyer M, Licht M, Atwood LW, Peoples M, Smith RG, Mortensen DA, & Koide RT (2016). Influence of pesticide seed treatments on rhizosphere fungal and bacterial communities and leaf fungal endophyte communities in maize and soybean. Applied Soil Ecology. doi: 10.1016/j.apsoil.2016.02.008
- Szink I, Davis EL, **Ricks KD**, & Koide RT (2016). New evidence for broad trophic status of leaf endophytic fungi of *Quercus gambelii*. Fungal Ecology. doi: 10.1016/j.funeco.2016.04.003

In Review

Ricks KD & Yannarell AC (In revision, ISME). Rooted in history: reviewing the contribution of plant-microbe eco-evolutionary dynamics in mediating host phenotypes.

Bedwell SL*, Lakis IM, Megow AR, **Ricks KD**, Lau JA, Fields CJ, Whitaker RJ, & Heath KD (In revision, ISME). Ecological shifts within *Rhizobium* communities limit plant health under long-term N-addition.

Ricks KD, Bhatt K*, & Frederickson ME (In revision, ISME). Microbial adaptation to environmental stress benefits stressed hosts.

In Prep

Ricks KD, Schwartz C⁺*, Blaszyński M*, Gonzalez D*, Heath KD, Lau JA & Yannarell AC (In prep, Target: PNAS). Symbiosis as engines for rapid adaptation: rhizobia evolution facilitates plant adaptation.

⁺Co-first authors

*Student mentee

AWARDS & FELLOWSHIPS

University of Toronto, Arts and Sciences Postdoctoral Fellowship	CA\$120,000	2023–2025
UIUC PEEC Summer Research Award	\$3,000	2019–2021
Harley J. Van Cleave Research Award	\$1,000	2019
UIUC PEEC Travel Grant	\$1,000	2019–2021
UIUC Graduate College Dissertation Grant	\$3,764	2019–2020
Illinois Distinguished Fellowship	\$75,000	2018–2021
BYU Department of Biology Travel Grant	\$700	2017

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant		
NRES 475 (Environmental Microbiology)-University of Illinois		2022–2023

- *Developed and delivered lectures for the course module on Microbial Symbiosis*

BIO 100 (Introductory Biology)-Brigham Young University

2014–2015, 2017

Undergraduate mentees

University of Toronto

Kasturi Bhatt

Makenna Brissette

University of Illinois

David Graves

Alexis Rhoades

Faith Smith

Chase Schwartz

Meghan Blaszyński

Gary Liu

Destiny Gonzalez

Yunqian Zhu

Jake Jagiello

Brigham Young University

Eliza Clark

Emily Aranda

Kate Harmer

PRESENTATIONS & POSTERS

Ricks KD & Frederickson ME (2026). Incorporating microbial evolution into plant community dynamics. Invited Conference talk: *Canadian Society for Ecology and Evolution*. Toronto, Canada.

Brissette M*, **Ricks KD** & Frederickson ME (2026). Investigating host, environmental, and microbial modulation of priority effects in the *Lemna* microbiome. Conference poster: *Canadian Society for Ecology and Evolution*. Toronto, Canada.

Ricks KD & Frederickson ME. (2025). Plant-microbe symbiosis mediates local adaptation in duckweed. Conference talk: *Evolution*. Athens, USA.

Bhatt K*, **Ricks KD**, Frederickson ME. (2024). Experimental evolution of nitrogen fixing bacteria under salt stress. Conference talk: *Global Change Science Symposium*. Toronto, Canada.

Schwarz C*, **Ricks KD**, Vereau Gorbitz D, Yannarell AC, & Heath KD. (2024). Standing variation or *de novo* mutation: What drives drought tolerance in experimentally evolved *Rhizobium leguminosarum*? Conference poster: *Lake Arrowhead Microbial Genomics*. Lake Arrowhead, USA.

Ricks KD, Heath KD, Blaszyński M, Gonzalez D, Yannarell AC. (2024). Symbiosis as engines for rapid adaptation: rhizobia evolution facilitates plant adaptation. Invited conference talk: *Canadian Society for Ecology and Evolution*. Vancouver, Canada.

Blaszyński M*, Gonzalez D, **Ricks KD**, & Heath KD. (2023). Swinging Bacteria: Impacts of *Rhizobium* on nonhost plant species (Medicago). Conference poster: *Genomics of eco-evolutionary multiscale symbioses*. Urbana, USA.

Gonzalez D*, Blaszyński M, **Ricks KD**, & Heath KD. (2023). Drought mitigating rhizobia. Conference talk: *Illinois Summer Research Symposium*. Urbana, USA.

- Ricks KD**, Heath KD, & Yannarell AC. (2022). Evolution of rhizobia populations alter plant adaptation to the moisture environment. Conference talk: Ecological Society of America. Montreal, Canada.
- Smith F*, **Ricks KD** & Yannarell AC. (2021). Biofilm production in rhizobia influences clover drought response. Conference talk: Illinois Summer Research Symposium. Urbana, USA.
- Ricks KD** & Yannarell AC. (2021). Plant adaptation to local environment through microbial interactions. Conference talk: Ecological Society of America. Virtual.
- Ricks KD** & Yannarell AC. (2020). Adaptive changes with rhizosphere microbial communities to historical watering conditions. Conference talk: Ecological Society of America, Virtual.
- Ricks KD** & Koide RT. (2019). Biotic filtering of endophytic fungal communities in *Bromus tectorum*. Conference talk: Ecological Society of America. Louisville, USA.
- Ricks KD**, Davis ER, & Koide R. (2017). Climate and dispersal influence the structure of leaf endophyte communities of *Quercus gambelii* in the eastern Great Basin. Conference talk: Ecological Society of America. Portland, USA.

*Student mentee

SERVICE

Department Service

UIUC Ecology and Evolution Graduate Admissions Committee	2021–2022
UIUC Ecology and Evolution Steering Committee	2021–2022

Reviewer

Science, National Science Foundation, Ecology, Evolution Letters, ISME, Proceedings of the Royal Society B, Annals of Botany, Functional Ecology, Microbial Ecology, Plant and Soil, Scientific Reports, Frontiers in Plant Science, Ecology and Evolution, Theoretical Ecology, Ecology, New Phytologist

Other

- Mentor for PRECS (Phenotypic Plasticity Research Experience for Community College Students) program at UIUC 2021
<https://precs.igb.illinois.edu/>
 Facilitated research experiences for community college students
- Mentor for Merit program at UIUC 2022–2023
<https://merit.illinois.edu/mpn/>
 Provided undergraduate students from underrepresented backgrounds mentorship in career and research development
- Participant in the 500 Queer Scientists Visibility Campaign